

MTEC2280 – Final Project Proposal

Title: Expanded Arpeggiated Musical Controller

Student: Jay Jung Gurung

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1. Concept Description & Inspiration

For my final project in MTEC2280, I will expand my earlier ESP32 ↔ p5.js musical interaction system into a more expressive, hardware-driven arpeggiated musical controller.

Previous project features:

- Play notes using the computer keyboard (sent from p5.js to ESP32)
- Switch between major and minor arpeggios
- Control tempo with a knob
- Hear sound through a piezo element

New version goals:

- Feature-rich, physical musical device
 - Added control and input methods
 - Improved audio
 - Custom fabrication
 - Small, self-contained controller blending physical interaction, serial communication, and sound generation on ESP32
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2. Sketch of the Design

Verbal layout:

- Compact control panel with:
 - Three rotary knobs (tempo, note length, root-note selector)
 - One slide potentiometer (volume)
 - Toggle switches for:
 - Major/Minor scale
 - Notes/Arpeggio
 - Sine/Square waveform
 - "Apply Root Note" button
 - Speaker with grill openings
 - ESP32 on custom PCB
 - All components in a custom 3D-printed enclosure
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3. Materials List

Hardware

- ESP32-S3 or ESP32-WROVER module
- Full speaker (8Ω) + audio amplifier (PAM8302 or PAM8403)
- 3 × rotary potentiometers (10kΩ): tempo, note length, root note scroll
- 1 × slide potentiometer (10kΩ) for volume
- 3 × toggle switches: Major/Minor, Notes/Arpeggio, Sine/Square waveform
- 1 × pushbutton (Apply Root Note)
- Custom PCB, JST connectors, wires, headers
- 3D printed enclosure (PLA/ABS)
- USB cable (serial communication)

Software

- Arduino IDE / PlatformIO (ESP32 programming)
- p5.js (keyboard input + visualization)
- KiCad (PCB design)
- Autodesk Fusion 360 (enclosure)

Physical Materials

- PLA filament
 - Screws (M3 or ¼-20)
 - Solder & fabrication tools
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4. Skills I Already Have

- Serial communication between p5.js and ESP32
 - Mapping analog inputs (potentiometers)
 - Digital input handling (buttons/toggles)
 - Generating square-wave tones with LEDC
 - Programming arpeggios and musical scales
 - p5.js input handling and UI
 - Basic PCB workflow and design
 - Basic 3D modeling and fabrication
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5. Skills I Need to Learn (Technical Research Challenge)

- Using DAC output on ESP32 for sine wave generation
- Audio amplifier integration with proper gain and wiring
- Designing, routing, and producing a manufacturable custom PCB
- Implementing note length envelopes that preserve BPM
- Expanding serial protocol to handle three rows of keyboard input

- Adding waveform switching logic in ESP32
 - Designing a professional enclosure with tolerances for screws, pots, and switches
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6. Precedent Research (with Short Explanations)

1. ESP32 Tone Generation Examples

[Random Nerd Tutorials: ESP32 PWM](#)

Shows method of generating tones using LEDC PWM. My project adds DAC, waveform selection, and arpeggiated logic.

2. DIY Arduino Synth Projects

[Notes and Volts](#)

Shows how physical knobs and switches shape sound. Mine uses ESP32 + p5.js serial communication + custom PCB.

3. KiCad Beginner PCB Guide

[KiCad Discover](#)

Helps with schematic → PCB → fabrication workflow. My PCB integrates microcontroller logic, UI controls, and amplifier system.

7. Week-by-Week Timeline

Week of 11/26 — Proposal Due

- Deliverable: Submit written proposal only.
- No build or coding required this week.

Focus this week:

- Turn in proposal
 - Begin light planning
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Week of 12/03 — WIP 1 Due

Hardware Goals

- Start breadboard prototyping
 - Test potentiometers (tempo, note length, root-note scroll)
 - Test toggle switches (major/minor, notes/arpeggio, sine/square)
 - Test Apply Root Note button
 - Begin basic wiring layout planning for PCB

Software Goals

- Expand p5.js to recognize three rows of keyboard input
- Define a compact serial protocol for expanded note data

- Update ESP32 sketch to accept:
 - Scale mode switch
 - Waveform mode
 - Note-length input
 - Root-note selection input

Fabrication Goals

- Begin first draft of PCB schematic in KiCad
- Start rough 3D enclosure planning (basic measurements)

Deliverables for WIP1

- Breadboard test of inputs
 - Updated p5.js note mapping
 - Early ESP32 code structure (pseudocode acceptable)
 - KiCad schematic draft (not routed)
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Week of 12/10 — WIP 2 Due

Hardware Goals

- Complete full breadboard prototype:
 - Tempo knob working
 - Note-length knob working
 - Root-note scroll + Apply button
 - Major/minor, sine/square, notes/arpeggio toggles
 - Speaker + amplifier integrated
 - Volume slider functional
 - Implement sine wave output via ESP32 DAC

Software Goals

- Implement note-length logic while maintaining BPM
- Integrate amplifier-safe amplitude control
- Update p5.js UI to reflect new modes and inputs
- Confirm all serial messages work with ESP32

Fabrication Goals

- Finalize full PCB layout (schematic + traces)
- Finalize 3D enclosure model in Fusion 360
- Order PCB from manufacturer

Deliverables for WIP2

- Fully functional breadboard prototype
- Working ESP32 + p5.js communication

- Ready-to-order PCB files (Gerbers)
 - Final 3D enclosure model render
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Week of 12/17 — Final Presentation & Critique

Hardware Goals

- Receive and assemble PCB
- Solder all components
- Mount PCB inside 3D-printed enclosure
- Install speaker, knobs, toggles, and secure all wiring
- Perform full system test & debugging

Software Goals

- Finalize ESP32 code (notes, arpeggios, waveform modes, BPM)
- Finalize p5.js interface & keyboard visual feedback
- Optimize serial timing

Final Deliverables

- Fully assembled device
 - Custom PCB
 - 3D-printed enclosure
 - Amplified speaker output
 - Full set of knobs + toggles working
- Demonstration video
- Final code
- Documentation with images and explanations
- In-class presentation